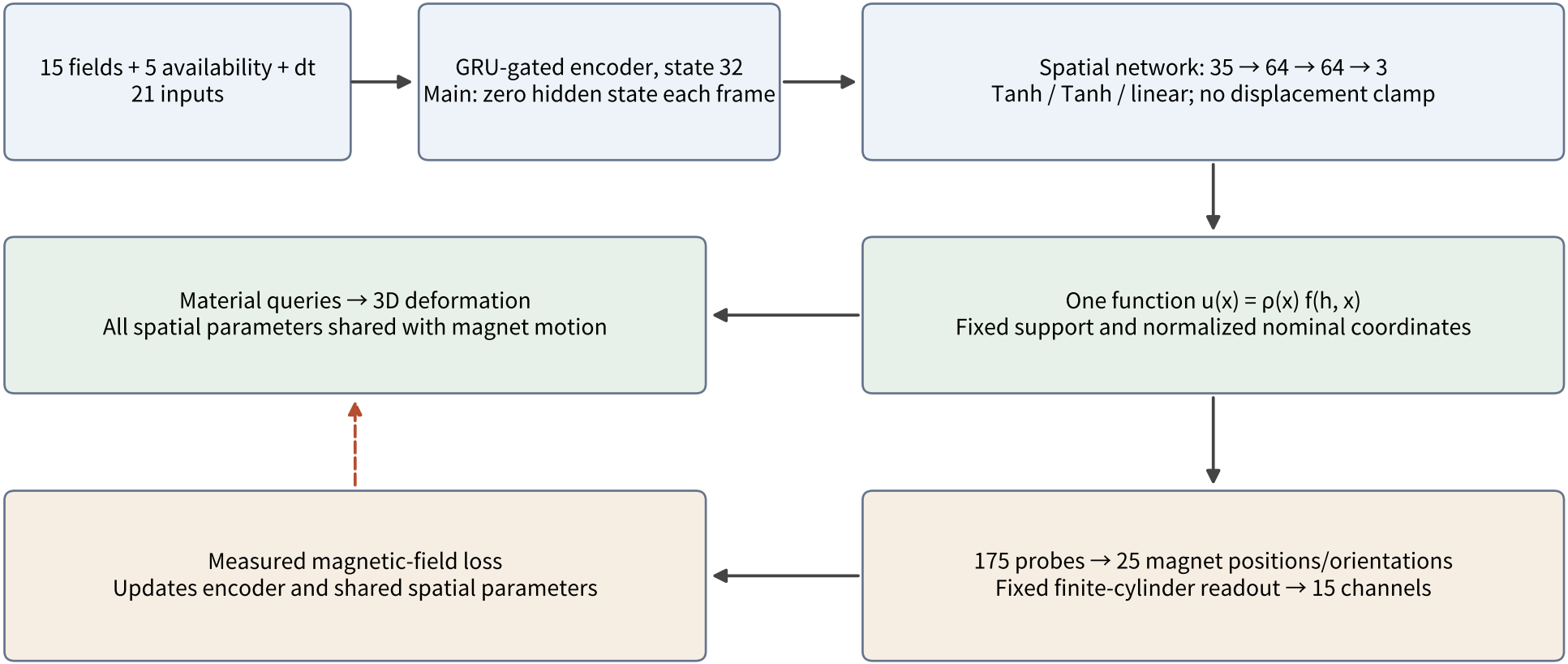Structural coupling exposes material states to magnetic-response supervision